ComplexCalc-HW — Power Supply Review

1 Purpose

This document summarizes the official power supply requirements of the STM32F411CEUx (datasheet DocID026289 Rev. 7, STMicroelectronics) for the **UFQFPN48** package, and compares them against what is currently routed in `design/ComplexCalc/ComplexCalc.kicad_sch` of the `UAA-F14/ComplexCalc-HW` repository. All reference figures and tables are direct captures from the official datasheet.

2 Actual pinout: UFQFPN48

Figure 1 (Figure 10 of the datasheet, page 34) shows the exact pin assignment of the package used on this board. From it we confirm:

- **3 VDD pins**: 24, 36, 48
- **3 VSS pins**: 23, 35, 47
- **Only 1 VCAP_1 pin** (pin 22) — this package does *not* bring out VCAP_2 (that signal only exists on the LQFP100 and UFBGA100 packages)
- VBAT: pin 1
- VDDA/VREF+: pin 9 · VSSA/VREF-: pin 8
- NRST: pin 7 · BOOT0: pin 44

Figure 10. STM32F411xC/xE UFQFPN48 pinout

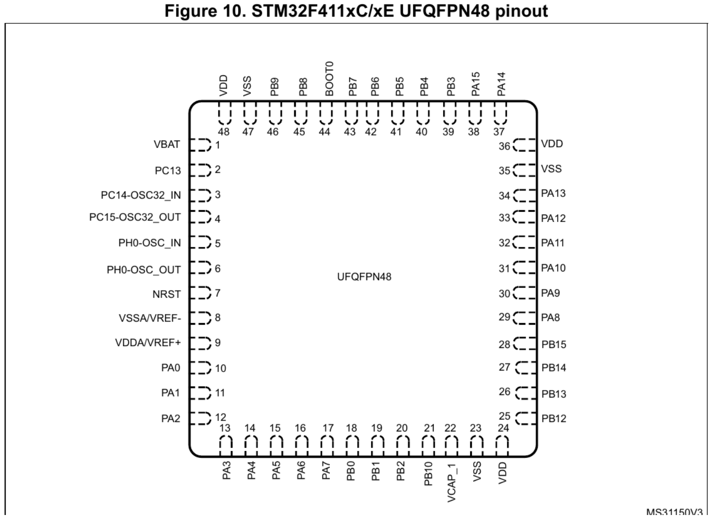


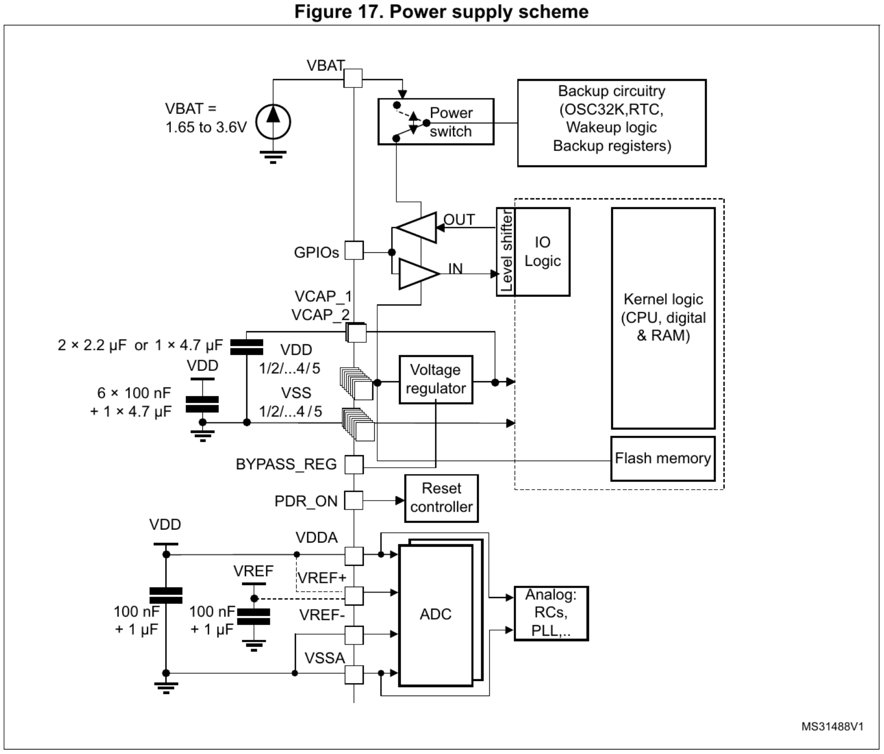
Figure 1: UFQFPN48 pinout — Figure 10, STM32F411xC/xE datasheet, p. 34

3 Official power supply scheme

Figure 2 (Figure 17, Section 6.1.6, page 59) is ST's reference scheme for decoupling each power supply branch.

6.1.6 Power supply scheme

Figure 17. Power supply scheme



1. To connect PDR_ON pin, refer to *Section 3.15: Power supply supervisor*.
2. The 4.7 μF ceramic capacitor must be connected to one of the V_{DD} pin.
3. VCAP_2 pad is only available on LQFP100 and UFBGA100 packages.
4. V_{DDA}=V_{DD} and V_{SSA}=V_{SS}.

Caution: Each power supply pair (for example V_{DD}/V_{SS}, V_{DDA}/V_{SSA}) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure good operation of the device. It is not recommended to remove filtering capacitors to reduce PCB size or cost. This might cause incorrect operation of the device.

Figure 2: Power supply scheme — Figure 17, Section 6.1.6, p. 59

4 VCAP capacitor for single-pin packages

Table 3 (Table 16, Section 6.3.2, page 65) gives the exact value that applies specifically when the package brings out only one VCAP pin (our case).

Table 16. VCAP_1/VCAP_2 operating conditions[1]

Symbol	Parameter	Conditions
CEXT	Capacitance of external capacitor with a single VCAP pin available	4.7 µF
ESR	ESR of external capacitor with a single VCAP pin available	< 1 Ω

1. When bypassing the voltage regulator, the two 2.2 µF V_{CAP} capacitors are not required and should be replaced by two 100 nF decoupling capacitors.

Figure 3: VCAP_1/VCAP_2 operating conditions — Table 16, Section 6.3.2, p. 65

5 Summary: required values per pin

Pin(s)	Required capacitor(s)	Source
VDD / VSS (pins 24/23, 36/35, 48/47)	100 nF per VDD/VSS pair (3 total) + 1×4.7 µF on any of the VDD pins	Figure 17, note 2
VCAP_1 (pin 22) → GND	**4.7 µF ceramic, ESR < 1 Ω** (single-VCAP package)	Table 16, Section 6.3.2
VDDA/VREF+ (pin 9) ↔ VSSA/VREF- (pin 8)	100 nF + 1 µF (single pair: on this package VDDA and VREF+ share a pin)	Figure 17
VBAT (pin 1)	100 nF, or a direct connection to VDD if no backup battery is used	Figure 17
NRST (pin 7)	Already has an internal pull-up — an external capacitor (100 nF, optional) only improves EMI immunity, it is not mandatory	Table 7 (NRST legend)
BOOT0 (pin 44)	No internal bias — external resistor (typ. 10 kΩ pull-down to boot from flash)	Section 3.13 "Boot modes"

6 Comparison against the actual schematic

Extracted from the `ComplexCalc.kicad_sch` netlist (exported with `kicad-cli`) on September 2, 2026.

Checked item	Required	In the schematic	Verdict
VCAP_1 (C14)	4.7 µF, ESR<1Ω	**2.2 µF**	**Below the required value** — 2.2 µF is the value for packages with *two* VCAP pins, not a single one.
VBAT (pin 1)	Tied to battery or to VDD	**Unconnected (isolated net)**	**Floating** — it is neither on the 3.3 V net nor does it have a capacitor.
VDD/VSS (3 pairs)	3×100 nF + 1 bulk	C10, C11, C12 (100 nF each) + C13 (1 µF) + C8 (22 µF)	Meets the requirement — bulk capacitance is even larger than the minimum required.
VDDA/VREF+ (pin 9)	100 nF + 1 µF local	Same 3.3 V net as VDD, with C13 (1 µF) present	Electrically compliant — physical proximity of the capacitor to the pin could not be verified from the schematic (check the layout).
NRST	Internal pull-up already sufficient; typ. cap. 100 nF	R4 (10 kΩ external, redundant) + C9 (**1 µF**, not 100 nF)	**Suggest lowering to 100 nF** — not an error (it is still functional), but 1 µF extends the reset time more than necessary. Confirmed against the actual BlackPill schematic (WeAct Studio): its reset circuit uses exactly **100 nF** at the same node.
BOOT0 (R5)	∼10 kΩ pull-down	10 kΩ	Correct
BOOT1/PB2 (R6)	Define level	10 kΩ pull	Correct
HSE crystal (Y1)	Must match `RCC.HSE_VALUE` in firmware	Y1 = **8 MHz** physically, but the `.ioc` has `HSE_VALUE=25000000`	**Review** — if the firmware ever enables HSE, the clock tree must be recalculated for 8 MHz, not 25 MHz.
Regulator (U2)	—	**AMS1117-3.3** (not NCP1117-3.3)	Documentation note: corrected in the repo's README.

7 Recommended actions

1. Change C14 from 2.2 µF to **4.7 µF** on VCAP_1.
2. Connect VBAT (pin 1) to the 3.3 V rail (or leave a 100 nF footprint if a real backup battery for the RTC is added in the future).
3. Lower C9 (NRST) from 1 µF to **100 nF** — the textbook value, and the one used by the actual reference BlackPill at the same node.
4. Before enabling HSE in firmware, correct `RCC.HSE_VALUE` to 8 MHz in the `.ioc` and regenerate the clock tree in STM32CubeMX.
5. Verify on the physical layout (not just the schematic) that the VDD/VDDA capacitors are placed as close as possible to their respective pins, since the bottom layer is already occupied by the matrix keypad.